

BCE loss function derivation

The Binary cross entropy is mathematically derived from the bernoulli probability distribution using the maximum likelihood estimation principle.

Step-1: Bernoulli distribution

for binary classification, the true label is

$$y \in \{0, 1\}$$

and the neural network predicts

$$a = p(y=1|x)$$

where a is the sigmoid output.

The bernoulli probability mass function is

$$p(y|a) = a^y (1-a)^{1-y}$$

Step-2: Likelihood function

Suppose we have N training samples:

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$$

Assuming all samples are independent,

$$L = \prod_{i=1}^N a_i^{y_i} (1-a_i)^{(1-y_i)}$$

this is called the likelihood.

Step-3: log-likelihood

taking algorithm: $\log L = \log \left(\prod_{i=1}^N \hat{a}_i^{y_i} (1 - \hat{a}_i)^{(1-y_i)} \right)$

using

$$\log(ab) = \log a + \log b$$

we get,

$$\log L = \sum_{i=1}^N \log \left(\hat{a}_i^{y_i} (1 - \hat{a}_i)^{(1-y_i)} \right)$$

using

$$\log(a^b) = b \log a$$

$$\log L = \sum_{i=1}^N \left[y_i \log(\hat{a}_i) + (1 - y_i) \log(1 - \hat{a}_i) \right]$$

Step-4: In ML, we want to maximize the log-likelihood

$$\max \log L$$

Instead of maximizing, optimizing algorithm usually minimize a loss function.

So, we take the negative: $\text{Loss} = -\log L$

therefore

$$Loss = - \sum_{i=1}^N [y_i \log(a_i) + (1-y_i) \log(1-a_i)]$$

Step 5: Average loss

Dividing by N :

$$L_{BCE} = - \frac{1}{N} \sum_{i=1}^N [y_i \log(a_i) + (1-y_i) \log(1-a_i)]$$

this is the binary cross entropy loss function.

single sample BCE

For one training example:

$$L = - [y \log(a) + (1-y) \log(1-a)]$$

where $a = \sigma(z) = \frac{1}{1+e^{-z}}$ = predicted probability

y = true label (0 or 1)

we differentiate L with respect to a .

Step 1: Differentiate both terms

$$\frac{dL}{da} = - \frac{d}{da} [y \log(a) + (1-y) \log(1-a)]$$

since y is constant.

$$\frac{dL}{da} = - \left[y \frac{d}{da} \log(a) + (1-y) \frac{d}{da} \log(1-a) \right]$$

using

$$\frac{d}{da} \log(a) = \frac{1}{a}$$

and by chain rule, $\frac{d}{da} \log(1-a) = \frac{1}{1-a} \cdot (-1) = -\frac{1}{1-a}$

substitute:

$$\frac{dL}{da} = - \left[\frac{y}{a} - \frac{1-y}{1-a} \right]$$

step-2: remove the outer minus sign

$$\frac{dL}{da} = -\frac{y}{a} + \frac{1-y}{1-a}$$

step-3: Take common denominator

$$\begin{aligned} \frac{dL}{da} &= \frac{a(1-y) - y(1-a)}{a(1-a)} \\ &= \frac{a - ay - y + ay}{a(1-a)} \\ &= \frac{a - y}{a(1-a)} \end{aligned}$$

in neural networks, if,

$$a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

then $\frac{\partial a}{\partial z} = a(1-a)$

using the chain rule:

$$\left[(1-u) \frac{\partial}{\partial b} (1-u) + \frac{\partial a}{\partial z} \frac{\partial}{\partial b} (1-u) \right] = \frac{\partial}{\partial b}$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z}$$

$$\frac{1}{n-1} = (1-u) \cdot \frac{1}{n-1} + \frac{a(1-u)}{a(1-u)} \cdot \frac{\partial}{\partial b}$$

$$= a - u$$

$$\left[\frac{1}{n-1} - \frac{1}{n} \right] = \frac{\partial}{\partial b}$$

remove the order minus sign

$$\frac{1}{n-1} + \frac{1}{n} = \frac{\partial}{\partial b}$$